# Qwen research: questions, interventions and evidence
H100 = red · H200 = green · arrows carry documented relations

2026-10-09

2026-10-09 · H200
Initial split topology
Jobs 22197, 22131, 22200, 22219
Compared early disaggregated layouts and repaired setup/protocol issues.
07 job 22197 reached 1,297.02 good output tokens/s at N32; 09 job 22131 reached 1,472.73 at N32 and job 22200 reached 1,649.27 at N40. Separate topology job 22219 reached 1,666.01 at N40. The layouts and loads differ; these are feasibility milestones, not a causal ablation.

motivated

2026-10-10

2026-10-10 · H200
Heterogeneous TP8 warmup failure
Jobs 22271
Workers loaded, but pair warmup timed out before benchmark scoring.
No score; localized the problem to setup/transfer path and motivated inspecting KV descriptor work.

derived

2026-10-10 · H200
Request-scoped NIXL descriptors
Jobs 22294
Built and coalesced descriptors only for request pages after 720 CPU equivalence cases.
Actual P/D warmup completed; N32 scored 1,422.96 good tokens/s. Establishes patched-path feasibility, not descriptor-only speedup.

motivated

2026-10-11

2026-10-11 · H200
Equal-TP recovery and preserved losses
Jobs 22750, 22778, 22809
Retained partial and timeout attempts; tested a separately scoped equal-TP8 descriptor route.
22809 reached 1,507.96 at N32 but only 95.38% joint SLA; N40/N48 failed eligibility. Not selected.

derived

2026-10-13

2026-10-13 · H200
Usable split baseline
Jobs 22415, 23102, 23104
Kept two TP4 prefill replicas feeding TP8/EP8 decode; established full runs and controls.
22415 delivered N48 at 1,912.30; later controls near 1,911. This became a matched-tuning baseline.

motivated

control

2026-10-15

2026-10-15 · H200
Round-robin prefill routing
Jobs 23325, 23607
Changed routing after poor first-token latency; followed with a fresh control.
Pilot 23325 scored 2,402.64 vs 2,075.11 control and lowered mean TTFT 1,437.63→333.25 ms; single observation, ITL approached limit.

motivated

2026-10-15 · H200
MNNVL plus Flashinfer interaction
Jobs 23778, 23839
Tested communication/attention paths and their combined pipeline with repeat evidence.
MNNVL reached 2,592.16 and repeat ~2,592.92, about 8% over matched N52 control; Flashinfer alone near tie. Combined configuration reached later 2,632.61.

2026-10-15 · H200
FP8 KV and decode DeepGEMM losses
Jobs 23608, 23609
Tested each backend on the routed pipeline at matched N52.
Both failed joint SLA; retained BF16 KV and closed those tested branches. Does not rule out phase-specific use.

2026-10-15 · H100
H100 platform becomes measurable
Jobs 220562.pbs111
Separated PBS launch and loader failures from a completed baseline.
First full N32 point ~1,565.16 after long startup; H100 performance remains a separate series.

motivated

2026-10-17

2026-10-17 · H200
DeepGEMM specifically in prefill
Jobs 24291, 24320, 24327
Changed prefill only after a decode-only DeepGEMM loss; held decode and workload, then repeated.
24291 reached 2,715.842; independent 24327 reached 2,716.787, +1.7401% vs fresh control 24320. H100 did not inherit this result.

derived

2026-10-18

2026-10-18 · H100
H100 HPC-Ops decode
Jobs 221263.pbs111, 221274.pbs111, 221275.pbs111
Compared decode-only HPC-Ops against controls and repeated the treatment.
2,308.308 vs 2,245.534 (+2.7955%); repeat 2,304.083 stayed above fresh control 221275. ITL improved, TTFT rose.

motivated

2026-10-19

2026-10-19 · H200
HPC-Ops decode with DeepGEMM/chunk8192
Jobs 24721, 24724
Revisited decode backend under changed prefill runner/chunk and compared against a same-load control.
N58 candidate 2,938.628 vs 2,780.403 (+5.6907%); TTFT rose, ITL improved. Promising interaction, not yet confirmed by this pair alone.

2026-10-19 · H200
Separate N60 repeat/control
Jobs 24722, 24732
Repeated the incumbent auto-decode configuration separately from the N58 treatment.
24732 2,836.531 vs 2,839.532 original control; confirms stability of that configuration only, not the N58 treatment.

2026-10-19 · H100
H100 both-phase HPC-Ops
Jobs 221315.pbs111, 221590.pbs111, 221645.pbs111
Added prefill HPC-Ops to the retained decode path and refined load with repeat controls.
Initial 2,371.961 vs 2,308.979 control; later N54 2,430.891 and repeat 2,418.588. N56 did not establish further gain.

control

2026-10-19 · H100
H100 prefill alternatives lose
Jobs 221828.pbs111, 221880.pbs111
Matched tests covered chunk6144, selectors and dense FlashInfer prefill.
Chunk6144 2,411.995 vs 2,413.802 control; dense FlashInfer 2,227.031 (−7.74%). Passing gates did not mean improvement.

derived

control

2026-10-20

2026-10-20 · H200
N60 HPC-Ops interaction repeat
Jobs 24912, 24913
Ran fresh matched controls and independently repeated the promising phase interaction.
Repeat improved 5.295% vs matched auto-decode control and was −0.033% below original treatment; mean TTFT/ITL tradeoff retained.

2026-10-20 · H100
H100 load/chunk frontier
Jobs 222281.pbs111, 222282.pbs111, 222383.pbs111
Measured chunk and load interactions with full-run latency gates.
N54 control 222281 scored 2,401.578; smaller chunk2048 lost −26.13%; 8192 and later H100 chunk branches did not establish a useful gain.

2026-10-20 · H100
H100 repeated frontier remains separate
Jobs 222048.pbs111, 222049.pbs111, 222050.pbs111
Reviewed three H100 full runs with independent observations.
2,433.020 / 2,395.311 / 2,406.802 good tokens/s; no pooling with H200 3,044 series.

motivated

2026-10-21

2026-10-21 · H200
N62 to N64 matched load
Jobs 25365, 25366
Tested a bounded concurrency increase on the revised recipe.
N62 3,011.875; N64 3,044.490 (+1.082867%) under full gates. This motivated a repeat, not a claim of certainty.

derived

2026-10-22

2026-10-22 · H200
Independent N64 repeat
Jobs 25373
Repeated the selected N64 configuration under the full protocol.
3,044.075, −0.013634% vs first N64; supported the historical repeated mean 3,044.400475.

motivated

2026-10-22 · H200
N64 versus N68 stop
Jobs 25487, 25488
Compared fresh full N64/N68 runs after the repeated N64 result.
Only +0.083056% at N68 while mean TTFT rose ~1,113.6→1,347.1 ms; retained N64 and closed N68.

control

2026-10-27

2026-10-27 · H200
Native full N64 reproduction
Jobs 26766
Replayed the retained native configuration with current gates.
3,045.501509 good output tokens/s; separate series from historical repeated mean.

motivated

2026-10-29

2026-10-29 · H200
Cold-prefill cache candidate
Jobs 27376, 27378
Compared a cache intervention to its own fresh control.
Candidate peak 3,048.446289 but only +0.240921%, below 1% gate; p95 TTFT worsened. No promotion.

2026-10-29 · H200
Tokenizer, prefill attention and EAGLE losses
Jobs 26949, 26951, 26953, 27077, 27238
Tested targeted backend/serving alternatives with same incumbent context.
Dynamic tokenizer gains +0.192/+0.270% not promoted; FlashInfer prefill −6.385%; shallow EAGLE3 −2.080%.

control

2026-10-30

2026-10-30 · H200
Sparse decode graph experiment
Jobs 27510
Reduced graph buckets to test memory proxy hypothesis.
3,029.861122 (−0.370208%); memory/rank fell 1.15–0.36 GB but throughput did not improve.

motivated

2026-10-31

2026-10-31 · H200
Direct H100-derived recipe check on H200
Jobs 27777, 27779, 27781, 27783
Fresh H200 control, FA3 decode and H100 recipe at N54/N64; preserved same-GPU comparison.
Control 3,038.022907; FA3 −1.276%; H100 recipe N64 −10.408%. Retained H200-specific recipe; no pooled GPU ranking.

motivated

2026-10-31 · H200
Phase-specific inference mode
Jobs 27826, 27828, 27830
Compared decode-only and prefill-only inference_mode ports with explicit no_grad control.
3,039.167 control; +0.006223% decode and +0.130765% prefill. Below 1% gate; no promotion.

motivated

2026-10-31 · H200
Refine transferred recipe chunk
Jobs 27804, 27806
Held HPC-Ops prefill runner and tested chunk6144/8192, then compared at shared chunk.
2,821.885 / 2,900.607; chunk8192 recovered vs 4096 but remained 4.523% below incumbent.

motivated

2026-11-02

2026-11-02 · H200
Exact packaged entry failure and acceptance
Jobs 28194, 28197
Fixed readiness check after health returned before model registration; reran exact entry.
28194 failed before benchmark; 28197 passed in 611 s at N64, 3,051.518173 good output tokens/s. Acceptance, not matched optimization.

motivated

2026-11-02 · H200
N16 quality pilot fails point threshold
Jobs 28367, 28368, 28371
Compared a bounded pilot before committing the full fixed study.
The first reference attempt 28367 failed; corrected reference 28371 scored 104 correct and candidate 28368 scored 102 (98.076923% retention). The pilot failed the local 99% point gate and motivated the full-set measurement.

motivated

2026-11-02 · H200
Complete paired local answer-quality test
Jobs 28373, 28375, 28382, 28384, 28386, 28388, 28390, 28392, 28394, 28396, 28398, 28400, 28402, 28404, 28406, 28408
Compared stock FP8 and accepted candidate on the same 12,032 MMLU-Pro questions, fixed prompts/settings/parser, in eight paired shards.
Reference 9,460/12,032 (78.623670%); candidate 9,452/12,032 (78.557181%); 99.915433% reference-score retention passes local 99% point gate. Not BTC certification or statistical equivalence.